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Polysemanticity and Capacity in Neural Networks

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Abstract

Individual neurons in neural networks often represent a mixture of unrelated features. This phenomenon, called polysemanticity, can make interpreting neural networks more difficult and so we aim to understand its causes. We propose doing so through the lens of feature *capacity*, which is the fractional dimension each feature consumes in the embedding space. We show that in a toy model the optimal capacity allocation tends to monosemantically represent the most important features, polysemantically represent less important features (in proportion to their impact on the loss), and entirely ignore the least important features. Polysemanticity is more prevalent when the inputs have higher kurtosis or sparsity and more prevalent in some architectures than others. Given an optimal allocation of capacity, we go on to study the geometry of the embedding space. We find a block-semi-orthogonal structure, with differing block sizes in different models, highlighting the impact of model architecture on the interpretability of its neurons.

1 Introduction

Individual neurons in neural networks often represent multiple unrelated features in the input [OMS17, OCS+20]. This phenomenon is known as polysemanticity, and makes it more difficult to interpret neural networks [OCS+20]. While "feature" is a somewhat fuzzy concept [EHO+22b], there are at least some cases where we "know it when we see it". For example, when the input features are independent random variables that do not interact in the data-generating process, neurons that represent combinations of these input features can be confidently called polysemantic. In this work we explore how loss functions incentivize polysemanticity in this setting, and the structure of the learned solutions.

Fittingly, there are multiple ways that polysemanticity can manifest. Here we focus on one form that seems particularly fundamental, namely superposition [EHO+22b]. Suppose we have a linear layer that embeds features which then pass through a layer with a nonlinear activation function. The feature embedding vectors might not be orthogonal, in which case multiple neurons (nonlinear units) are involved in representing each feature. When there are at least as many features as neurons this means that some neurons represent multiple features, and so are polysemantic (Figure 1, right). There are other causes of polysemanticity, e.g. feature embedding vectors could be rotated relative to the neuron basis (Figure 1, left), but we do not study these in this work.

Here we build on the work of [EHO+22b], who studied polysemanticity in the context of toy models of autoencoders. They found that models can support both monosemantic and polysemantic neurons, that polysemantic neurons can perform certain kinds of computations, and that the embedding vectors of features often formed repeating motifs of a few features symmetrically embedded in a low-dimensional subspace. Moreover, in their models they found distinct "phases" where superposition was either significant or completely absent. Sparser inputs resulted in more superposition. Features with similar importance were more likely to be in superposition. This reflects an abundance of unexpected structure, and gives new handles on the phenomenon of polysemanticity.

We study these phenomena through the lens of *capacity*, or the fraction of an embedding dimension allocated to each feature (Section 2, also termed "dimensionality" by [EHO+22b]). This ranges from 0-1 for each

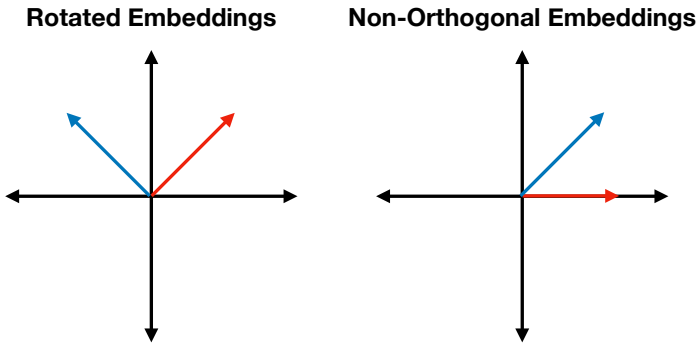

Figure 1: Feature embedding vectors are shown in two dimensions. The neuron basis corresponds to the coordinate axes. Left: rotated embeddings. Right: non-orthogonal embeddings. In both cases the result is polysemanticity because each neuron receives some input when either feature is present.

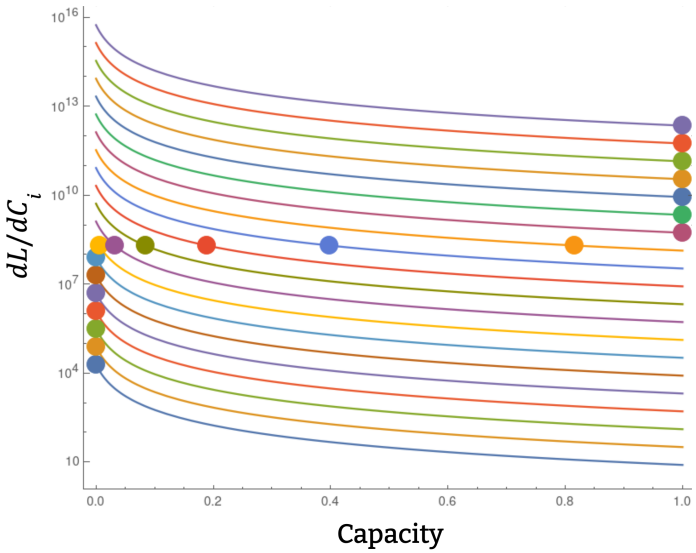

Figure 2: The marginal loss reduction $-\partial L/\partial C_i$ is shown for several features as a function of feature capacity in our toy model. Circles represent the optimal capacity allocation for a particular total embedding dimension. Colors vary to make individual curves more distinguishable.

feature, and the total capacity across all features is bounded by the dimension of the embedding space. Because the model has a limited number of neurons and so a limited number of embedding dimensions, there is a trade-off between representing different features. We find that the capacity constraint on individual features (0-1) means that many features are either ignored altogether (not embedded) or else allocated a full dimension orthogonal to all the other features in the embedding space, depending on the relative importance of each feature to the loss. Features are represented polysemantically only when the marginal loss reduction of assigning more capacity to each is equal (Figure 2). This neatly explains the sharp "pinning" of features to either 0 or 1 capacity noted by [EHO+22b], and gives us a framework for understanding the circumstances under which features are represented polysemantically.

To explore capacity allocation in a concrete model, we instantiate our theory for a one-layer model with quadratic activations (Section 3). Our model differs from the Anthropic toy model in that ours uses a different activation function to make the math more tractable, and, more importantly, ours is focused on polysemantic computation rather than data compression. We contrast these toy models in Figure 3.

For our toy model we can analytically determine the capacity allocation as a function of feature sparsity and importance (i.e. weight in the loss), and so construct a "phase diagram" (Figure 4). While the details of our phase diagram differ from those of [EHO+22b], reflecting our different toy model, there are three qualitative features that are in good agreement. First, when a feature is much more important than the rest,

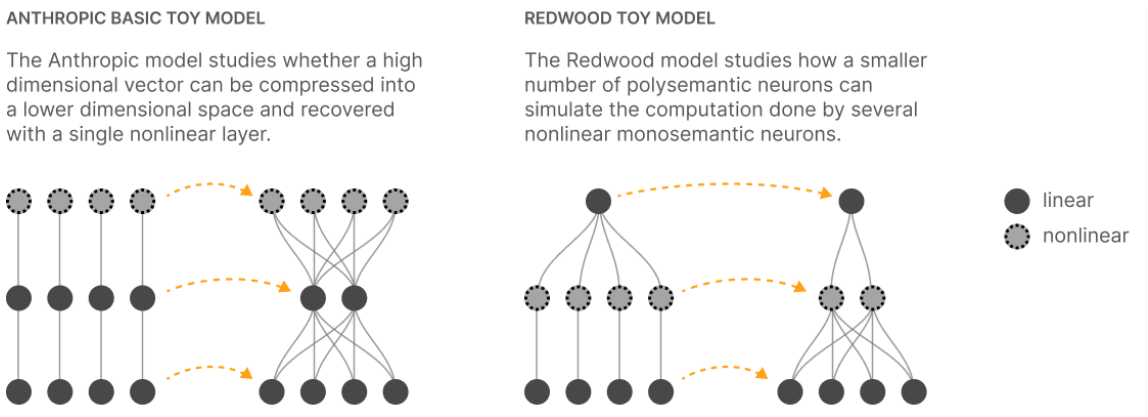

Figure 3: Comparison between the Anthropic toy model of [EHO+22b] (left) and our toy model (right). Model inputs are at the bottom of the diagram and outputs are at the top. The key difference is that the Anthropic model studies the compression and recovery of high-dimensional vectors, while ours examines how a smaller number of polysemantic neurons can simulate the computation done by a larger number of monosemantic ones. Figure kindly provided by Chris Olah.

it is always represented fully with its own embedding dimension. Second, when a feature is much less important than the rest, it is ignored entirely. Finally, in a sparsity-dependent intermediate region features are partially represented, sharing embedding dimensions. In addition, this confirms our theoretical expectation that capacity is allocated according to how much each feature matters to the loss (a mixture of importance and sparsity) and that it is often allocated to fully ignore some features while fully representing others. We supplement this with empirical results for a variety of activation functions showing that the phase diagram predicts the behavior of a broad family of 2-layer models.

We then turn to study the geometry of the embedding space (Section 4). When embedding matrices fully utilize the available capacity we call them "efficient". We find that every efficient embedding matrix has a block-semi-orthogonal structure, with features partitioned into different blocks. When multiple features in a block are present they interfere with each other, causing spurious correlations in the output and hence greater loss. Features do not, however, interfere across blocks.

The blocks in efficient matrices correspond to the polytope structure [EHO+22b] found, with small blocks corresponding to features embedded as regular polytopes and large blocks corresponding to less-ordered structures. Large- and small-block arrangements come with different advantages. With large blocks there is significant freedom to allocate capacity across features, whereas with small blocks there is the additional constraint that the capacity of each block be an integer and that the block capacities add up to the total capacity. On the other hand, with small blocks the lengths of embedding vectors can be chosen more freely because blocks can be scaled independently of each other without affecting the capacity allocation.

In our quadratic model the embedding matrices in our toy model always have one large block, which is correspondingly less structured. We expect that differences in architecture can lead to different sizes of blocks, which could provide a way to control the extent of polysemanticity in models, alongside other approaches such as changing the activation function [EHO+22a].

2 Capacity and Superposition

2.1 Definitions

Suppose we have a model composed of stacks of linear layers with nonlinear activation functions. In each layer, the model applies a linear transform to the input vector x to produce an embedding vector e, and then performs an element-wise non-linear calculation on those embeddings to produce the non-linear activation vector h. For instance, we might have